

SAP CPI Project Documentation

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Scenario: Idempotent Process Call IFlow in SAP Integration Suite

1.Scenario Overview

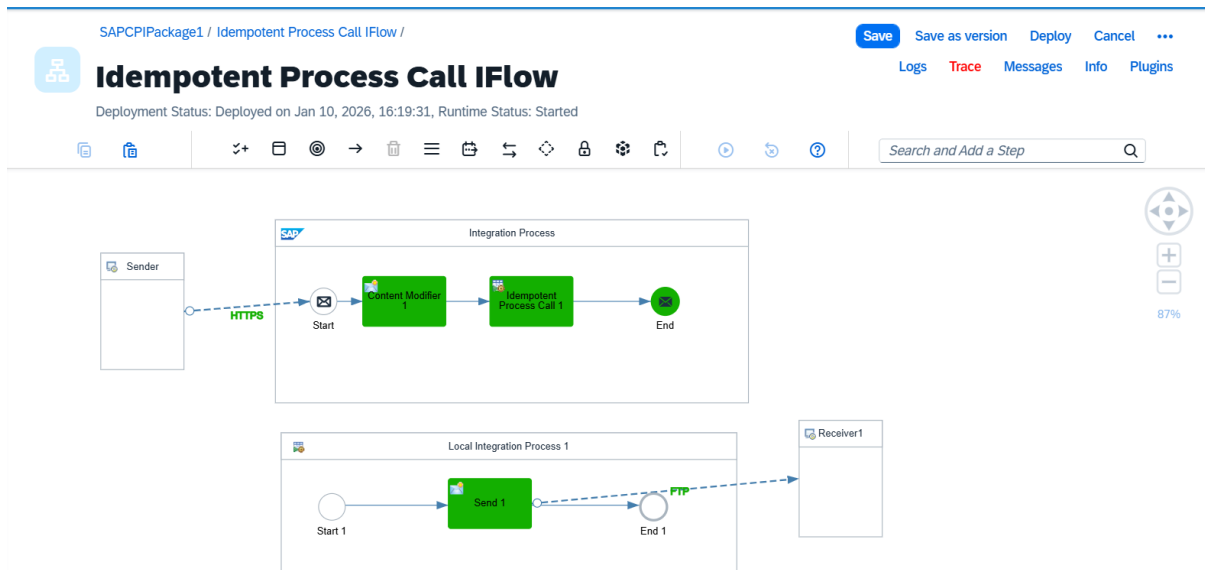
In cases where the **backend system is non-idempotent**, it cannot detect or manage duplicate messages on its own. Or if the HTTP method used is non-idempotent. That's where the **middleware** must take responsibility.

- a **unique ID** is required to identify duplicates. In our scenario, the unique reference is the **Sales Order ID** included in the order XML.
- This ID serves as the key for the middleware to check whether a message has already been processed. If the middleware encounters this Sales Order ID for the first time, it allows the message to pass through to the backend system.
- If the same ID appears again, whether due to network retries, system errors, or users clicking "Submit" multiple times, the middleware can safely **ignore the duplicate message**.
- By implementing this check, the middleware ensures that the backend system receives each order **only once**, even if the message has been sent multiple times.

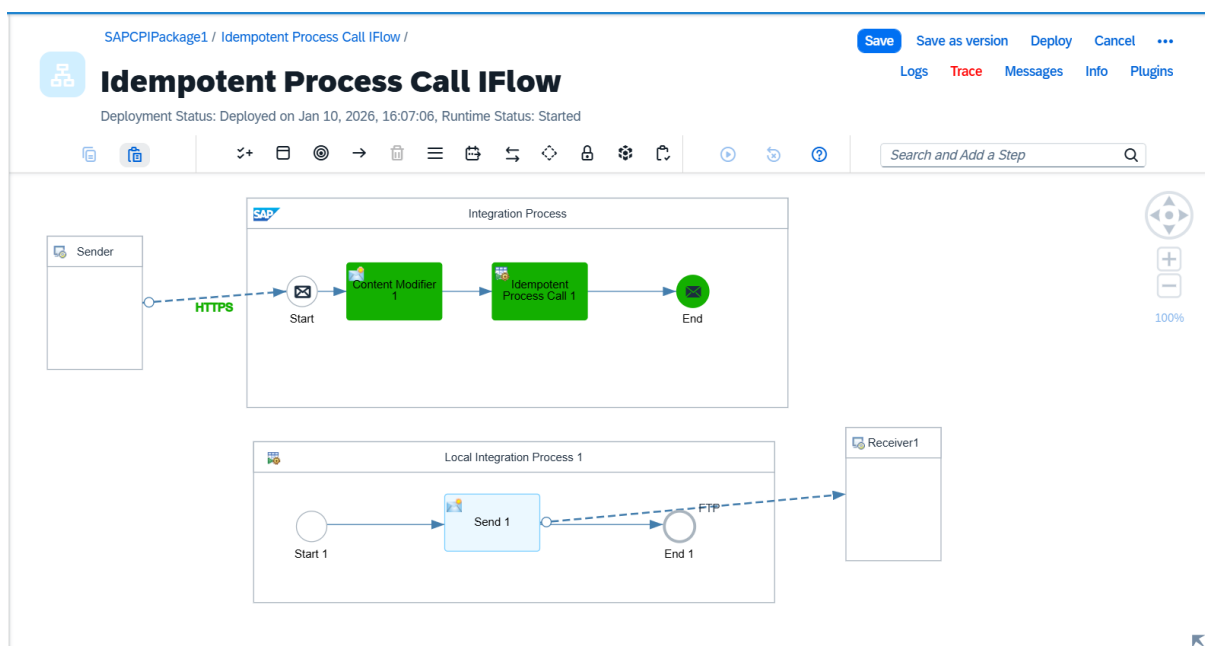
Prerequisites for Using the Idempotent Process Call in SAP Integration Suite

- One of the prerequisites for using the Idempotent Process Call in SAP Integration Suite is that we must have a unique ID to identify each message (or iteration).
- This ID is then used to detect duplicates and ignore any subsequent duplicate messages associated with that same ID.

2. Architecture Diagram



Message Processed successfully



Local process call skipped

3. Step-by-Step Implementation

- Incoming sales order messages are received and processed in the main integration flow.
- Extracted the Order ID from the payload using an XPath expression to uniquely identify each message.

- Configured an Idempotent Process Call using the extracted Message ID to ensure duplicate message handling.
- Enabled “Skip Process Call for Duplicates” option so that duplicate messages are automatically ignored by the system.
- Implemented a Local Integration Process responsible for forwarding valid and unique messages to the FTP receiver system.

Successful Test case: Message Processed successfully

- Tested the interface by sending a unique sales order, which was successfully processed and received by the FTP target system.
- Verified message processing using SAP Integration Suite Message Monitoring and Traces.

Failed Test case: Local Process call skipped

- Sent a duplicate sales order message and confirmed that the FTP receiver did not receive the message.
- Observed in the Message Processing Log that the Local Integration Process Call was skipped for the duplicate message.

4.Adapter Configuration:

HTTPS Sender Adapter

Endpoint: <https://b84a8864trial.it-cpitrial05-rt.cfapps.us10-001.hana.ondemand.com/http/idempotentCall>

Method: POST

Authentication: Basic

FTP Receiver Adapter

The screenshot shows the SAP Integration Suite configuration for an Idempotent Process Call IFlow. The interface includes a top navigation bar with 'Edit', 'Configure', and 'Deploy' buttons. Below the title 'Idempotent Process Call IFlow', there is a status bar indicating 'Unsaved Changes' and 'Runtime Status: Started'. The main configuration area is divided into three tabs: 'General', 'Target', and 'Processing'. The 'Target' tab is currently selected, showing the following parameters:

- FILE ACCESS PARAMETERS:**
 - Directory: UniqueFiles1
 - File Name: \${property.OrderId}
 - Append Timestamp: ☒
- CONNECTION PARAMETERS:**
 - Address: ftp.drivehq.com
 - Proxy Type: Internet
 - Encryption: Explicit FTPS
 - Credential Name: FTP_Connect2
 - Timeout (in ms): 10000
 - Maximum Reconnect Attempts: 3

5.Integration Components:

Content Modifier:

The screenshot displays the SAP CPI Studio interface for an integration process named "Idempotent Process Call IFlow". The process flow is visible in the top pane, showing a sequence: Sender (HTTP) -> Start -> Content Modifier 1 -> Idempotent Process Call 1 -> End. The bottom pane is focused on the "Content Modifier" component, with the "Exchange Property" tab selected. Below the tabs, a table defines the properties for the "Create" action.

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	OrderId	XPath	//OrderId	java.lang.String	

Extract Unique ID from incoming payload

Idempotent Process Call:

The screenshot shows the SAP CPI Studio interface for the same "Idempotent Process Call IFlow". The process flow is identical to the previous screenshot. The bottom pane is now focused on the "Idempotent Process Call" component, with the "Processing" tab selected. This tab shows the configuration for the idempotent condition, including the message ID and a checkbox for skipping duplicates.

Local Integration Process: Local Integration Process 1

IDEMPOTENT CONDITION DETAILS

Message ID: `${property.OrderId}`

Skip Process Call for Duplicates: ☒

6. Input XML

```
<?xml version="1.0" encoding="UTF-8" ?>
```

```
<Order>
```

```
<OrderID>10250</OrderID>
<CustomerID>VINET</CustomerID>
<EmployeeID>5</EmployeeID>
<OrderDate>1996-07-04T00:00:00Z</OrderDate>
<RequiredDate>1996-08-01T00:00:00Z</RequiredDate>
<ShippedDate>1996-07-16T00:00:00Z</ShippedDate>
<ShipVia>3</ShipVia>
<Freight>32.38</Freight>
<ShipName>Vins et alcools Chevalier</ShipName>
<ShipAddress>59 rue de l'Abbaye</ShipAddress>
<ShipCity>Reims</ShipCity>
<ShipRegion></ShipRegion>
<ShipPostalCode>51100</ShipPostalCode> <ShipCountry>France</ShipCountry>
</Order>
```

Idempotent Repository and Validity Period

- However, note that the idempotent repository data is retained for **90 days only**. After 90 days, the stored Message IDs are removed, meaning duplicate checks can only be performed within a 90-day window from the first message execution. In most real-world scenarios, this time frame is more than sufficient